

Asset-Class Portfolio Optimization Using Gurobi

A mixed-integer mean-variance model with fixed transaction costs

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Abstract

This paper develops and evaluates an asset-class allocation model for a manufacturing company seeking to invest \$1,000,000 in surplus funds for one year. The investment universe contains U.S. Large-Cap Equity, U.S. Small-Cap Equity, International Equity, Corporate Bonds, and Treasury Bonds. Twelve monthly observations are used to estimate annual expected returns and an annualized sample covariance matrix. A mixed-integer quadratic optimization model then maximizes expected return while penalizing portfolio variance and fixed transaction costs. Binary variables represent whether an asset class is selected, and continuous variables determine portfolio weights. A minimum allocation of 5% is imposed on every selected asset. With a risk-aversion coefficient of 20, the model selects U.S. Large-Cap Equity and Treasury Bonds, assigning 50.45% and 49.55%, respectively. The resulting portfolio has an estimated annual return of 8.59%, annualized standard deviation of 3.05%, and fixed transaction costs of \$3,000. The result illustrates how a high risk penalty can favor a near-balanced combination of growth and defensive assets.

Keywords: *portfolio optimization, mean-variance analysis, mixed-integer quadratic programming, Gurobi, transaction costs, asset allocation*

1. Introduction

Manufacturing companies occasionally hold surplus cash that is not immediately required for production, payroll, or capital expenditure. Investing these funds can generate additional income, but a decision based only on the asset with the largest historical return may expose the company to unacceptable volatility. The company must therefore consider expected return, the risk of individual assets, how assets move together, and the cost of opening positions.

The present problem concerns the allocation of \$1,000,000 across five broad asset classes for a one-year planning horizon. The model is based on the mean-variance framework, in which expected return represents reward and covariance-based portfolio variance represents risk. The decision is made more realistic by adding fixed transaction costs and a binary selection decision for each asset class. These features transform the problem into a mixed-integer quadratic program that can be solved with Gurobi.

2. Problem Statement

The company may invest in U.S. Large-Cap Equity, U.S. Small-Cap Equity, International Equity, Corporate Bonds, and Treasury Bonds. If an asset class is selected, at least 5% of the portfolio must be allocated to it. No money may be assigned to an unselected class. Selecting an asset triggers a fixed fee: \$2,000 for U.S. Large Cap, \$2,000 for U.S. Small Cap, \$2,500 for International Equity, \$1,500 for Corporate Bonds, and \$1,000 for Treasury Bonds.

The optimization seeks the allocation offering the best modeled tradeoff among expected annual return, portfolio variance, and transaction-cost penalties. Risk aversion is set to 20, making risk reduction substantially more important than it would be in a return-oriented formulation with a small coefficient.

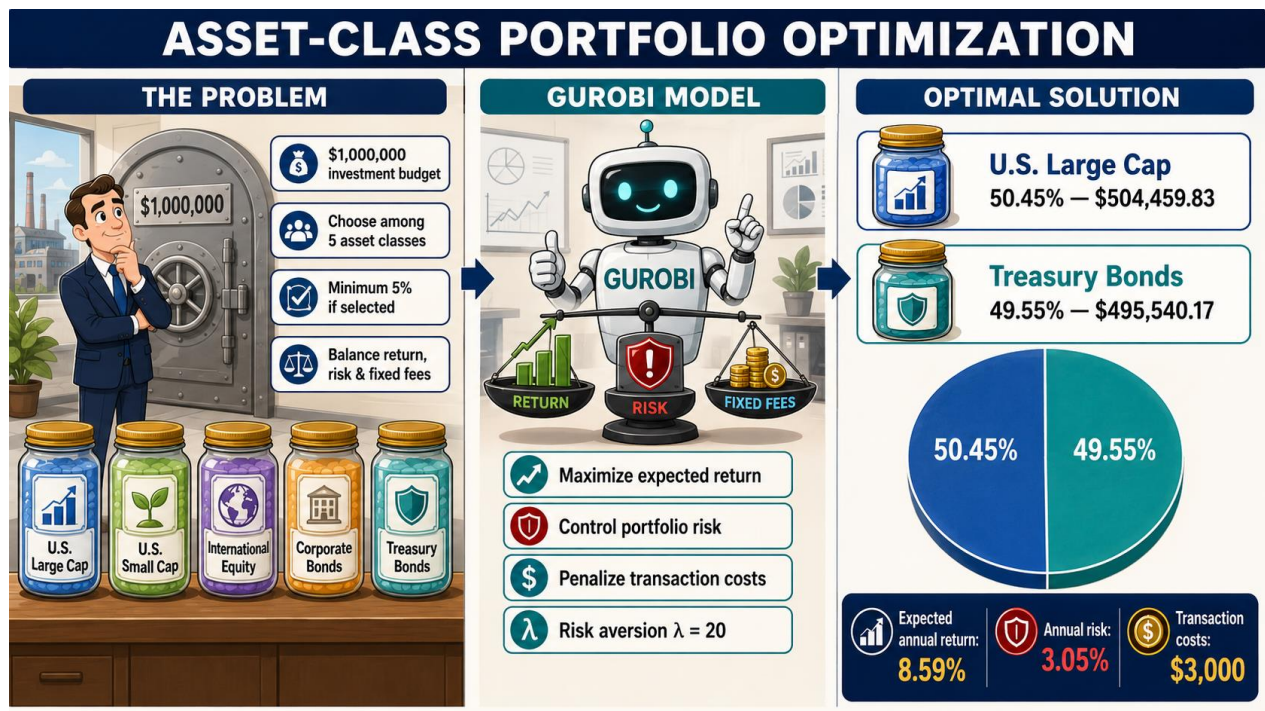


Figure 1. Cartoon summary of the investment problem, Gurobi model, and optimal solution.

3. Data Preparation and Parameter Estimation

The supplied spreadsheet contains 12 monthly return observations for each asset class. After removing the Month column, the program standardizes column names by replacing spaces with underscores and converts all remaining values to numeric form. If the spreadsheet stores percentages as Excel percentage values, a displayed value of 2.1% is read by pandas as 0.021, which is the correct decimal representation for the calculations.

The expected annual return for asset i is estimated by multiplying its arithmetic average monthly return by 12. The monthly sample covariance matrix is also multiplied by 12 to obtain an annualized covariance matrix. These annualization rules assume that monthly returns are comparable across the sample and that the one-year covariance can be approximated by linear time scaling.

$$\mu_i = 12 \times \text{mean monthly return of asset } i$$

$$\Sigma = 12 \times \text{sample covariance matrix of monthly returns}$$

Asset class	Annual expected return	Fixed cost
U.S. Large Cap	12.70%	\$2,000
U.S. Small Cap	15.80%	\$2,000
International Equity	8.90%	\$2,500
Corporate Bonds	5.10%	\$1,500
Treasury Bonds	4.40%	\$1,000

4. Mathematical Formulation

Let i and j index the five asset classes. The model uses two decision variables:

- w_i : fraction of the investment budget allocated to asset i .
- y_i : binary variable equal to 1 if asset i is selected and 0 otherwise.

4.1 Objective Function

The objective maximizes expected portfolio return after subtracting a risk penalty and normalized fixed transaction costs. Here, λ is the risk-aversion coefficient, μ_i is the annual expected return, σ_{ij} is the annualized covariance between assets i and j , c_i is the fixed transaction cost, and B is the \$1,000,000 budget.

$$\text{Maximize } \sum_i \mu_i w_i - \lambda \sum_i \sum_j \sigma_{ij} w_i w_j - \sum_i (c_i/B) y_i$$

4.2 Constraints

The weights must sum to one, meaning the code invests the full \$1,000,000:

$$\sum_i w_i = 1$$

The linking constraint prevents investment unless the corresponding asset is selected:

$$w_i \leq y_i \text{ for every asset } i$$

The minimum-allocation constraint requires at least 5% in each selected asset:

$$w_i \geq 0.05 y_i \text{ for every asset } i$$
$$0 \leq w_i \leq 1; \quad y_i \in \{0,1\}$$

5. Optimization Results

Gurobi selects only U.S. Large-Cap Equity and Treasury Bonds. The optimal portfolio assigns 50.445983% to Large Cap and 49.554017% to Treasury Bonds. The other three asset classes receive zero allocation and their binary selection variables equal zero.

Asset class	Selected	Weight	Amount
U.S. Large Cap	Yes	50.45%	\$504,459.83
U.S. Small Cap	No	0.00%	\$0.00
International Equity	No	0.00%	\$0.00
Corporate Bonds	No	0.00%	\$0.00
Treasury Bonds	Yes	49.55%	\$495,540.17
Total investment	—	100.00%	\$1,000,000.00

5.1 Return, Risk, and Objective Value

The expected annual portfolio return is 0.085870, or 8.59%. Applied to the \$1,000,000 investment, this corresponds to an expected annual gain of \$85,870.17 before transaction costs. Portfolio variance is 0.000931575, giving an annualized standard deviation of 3.0522%. The two selected assets incur \$3,000 in fixed transaction costs.

Performance measure	Value
Expected annual return	8.5870%
Expected annual gain before fees	\$85,870.17
Portfolio variance	0.000931575
Annualized standard deviation	3.0522%
Fixed transaction costs	\$3,000.00
Expected gain after fees	\$82,870.17
Mean-variance objective value	0.06423866

The objective value is obtained as $0.08587017 - 20(0.000931575) - 0.003 = 0.06423866$. This value is a utility score used to compare feasible portfolios; it is not the portfolio's expected return. The risk term reduces the score by approximately 1.8632 percentage points, while the normalized fee term reduces it by 0.30 percentage points.

6. Discussion

U.S. Small-Cap Equity has the highest estimated annual return at 15.80%, but it also has the largest variance and strong positive covariance with the other equity classes. With $\lambda = 20$, its additional expected return is not sufficient to compensate for its contribution to portfolio risk. Treasury Bonds have the lowest expected return but extremely low variance and negative covariance with U.S. Large-Cap Equity. Combining the two assets therefore produces diversification: the equity component provides growth potential, while Treasury Bonds stabilize the portfolio.

The near-even allocation is not imposed by the model. It emerges from the estimated return and covariance values together with the selected risk-aversion coefficient. A smaller λ would place greater emphasis on return and could produce a portfolio concentrated in equities. A larger λ would shift the allocation further toward low-risk assets. The result should therefore be interpreted as the optimal portfolio for the stated data, fee assumptions, and $\lambda = 20$ —not as a universally optimal investment strategy.

7. Modeling Limitation and Recommended Correction

The current code requires the portfolio weights to sum to 1 and then penalizes fees only in the objective. Consequently, it invests the full \$1,000,000 and treats the \$3,000 fees as an additional economic cost. If fees must be paid from the same \$1,000,000, the budget constraint should include them directly.

A budget constraint that deducts fixed fees from available funds is:

$$\sum_i Bw_i + \sum_i c_i y_i \leq B$$

Under this corrected treatment, the amounts invested would sum to \$997,000 when Large Cap and Treasury Bonds are selected, with the remaining \$3,000 paying their transaction costs. Other limitations include reliance on only 12 historical observations, the assumption that arithmetic monthly averages scale linearly, and the use of variance as the sole risk measure. A production model could use a longer history, scenario analysis, rebalancing constraints, liquidity requirements, and sensitivity analysis across multiple λ values.

8. Conclusion

This project demonstrates how predictive information derived from historical returns can be integrated with prescriptive optimization. Pandas estimates annual returns and covariance from observed monthly data, while Gurobi determines which asset classes to select and how much to allocate. For a \$1,000,000 budget and risk aversion of 20, the implemented model recommends investing \$504,459.83 in U.S. Large-Cap Equity and \$495,540.17 in Treasury Bonds. The portfolio is expected to earn 8.59% annually with 3.05% annualized volatility and \$3,000 in fixed transaction costs. The solution illustrates the practical value of optimization: rather than simply selecting the asset with the highest historical return, the model identifies a diversified allocation that reflects return, joint risk, selection requirements, and costs.

References

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